GHOSTNET — COMPLETE PROJECT HANDOFF REPORT

AI-Powered Moving Target Defense for Hospital IoT + Cloud Security

Student: SANGAVI | Machine: Windows 11 | Python: 3.12.7

Project Path: C:\Users\SANGAVI\Documents\project\ghostnet

Report Date: June 2026

SECTION 1 — WHAT THIS PROJECT IS

One Line

GhostNet is an AI agent that continuously changes the shape of a hospital network so attackers who map it today find a completely different network tomorrow.

Full Description

GhostNet is a Reinforcement Learning-driven Moving Target Defense (RL-MTD) system that protects the ICU infusion pump to cloud pharmacy command pipeline in hospitals. It deploys a single PPO agent that simultaneously mutates cloud VPC infrastructure and IoT edge gateway configurations in response to live threat intelligence, without disrupting active patient care.

The Problem It Solves

In November 2022, AIIMS Delhi was ransomwared for 15 days. 40 million patient records exposed. Attackers spent weeks silently mapping a static network before striking. No existing security product stops the mapping phase. GhostNet eliminates the static network permanently.

Domain

Hospital — specifically the ICU infusion pump to cloud pharmacy API pipeline.

Why Hospital Over Navy

Hospital is public, demonstrable, no security clearance needed, directly tied to AIIMS Delhi real incident, and companies like Apollo, Narayana, Philips Healthcare are actively spending on this problem right now.

SECTION 2 — THE 5 CORE INNOVATIONS

Name	Full Form	What It Does	Why It Is New
DUAS	Dual-Domain Unified Action Space	Single RL agent controls BOTH cloud VPC and IoT gateway mutations in one MDP	No existing paper has a unified action space spanning both domains
AESM	API Endpoint Schema Mutation	Rotates REST API URL paths, parameter names — not just IPs and ports	No MTD paper has ever mutated the application layer
TADR	Traffic-Aware Dual-Objective Reward	Reward penalises both attacker disruption AND legitimate traffic disruption	First MTD reward function that explicitly protects patient care
LTSA	Live Threat-Feed State Augmentation	Real NIST CVE + Shodan scores injected into RL state vector	Agent reacts to real-world published vulnerabilities, not just local scans
ARAC	Adversarial Red Agent Co-training	CALDERA red agent co-evolves with defender in self-play	First MTD evaluated against adaptive adversarial RL opponent

All 5 are ONE project. Not separate. DUAS is the engine. LTSA and AESM are inputs. TADR is the reward brain. ARAC is the evaluation method.

SECTION 3 — RESEARCH GAP (what existing papers miss)

Existing Paper	What It Does	What It Misses
Gayathri et al. VIT Chennai, Sensors 2023	MTD on AWS for IoT cloud security	Rule-based mutation, no RL, no API schema mutation, cloud only
Sharma, U of Toronto, Electronics May 2025	MTD evaluation using time-to-compromise	IoT only, random shuffling, calls RL-MTD "open future work"
Palo Alto Networks Medical IoT Security 2024	Zero Trust for medical devices	Reactive detection, no MTD, no IoT-cloud bridge protection
Claroty xDome Healthcare	Device visibility and anomaly detection	Static network, no mutation capability

Existing Paper	What It Does	What It Misses
Armis Cybersecurity Blueprint 2025	IoT and OT asset intelligence	Passive monitoring only, acknowledges gap but no solution

The University of Toronto 2025 paper explicitly says "future work: context-aware, threat-intelligent MTD." GhostNet is the answer to that open problem.

SECTION 4 — TECH STACK (complete)

Component	Technology	Purpose
RL Algorithm	PPO — stable-baselines3	Agent training
Environment	Custom Gymnasium env	Hospital network simulator
Cloud mutations	AWS boto3 + Terraform CDK	Real VPC security group changes (Phase 3)
IoT mutations	OpenWRT UCI API + paramiko	Gateway IP, firewall, MQTT rotation (Phase 4)
API mutations	AWS API Gateway / Kong	REST endpoint schema rotation (AESM)
Threat intel	NIST NVD CVE API (free)	Live CVE scores into state (working now)
Threat intel	Shodan API (academic free)	Internet exposure score (simulated now)
State encoder	PyTorch LSTM	Sequential threat feed processing
Red agent	MITRE CALDERA	Adversarial attack simulation (Phase 5)
Monitoring	Grafana + Prometheus	Live mutation dashboard (Phase 5)
Logging	ELK Stack	Audit trail
IoT hardware	Raspberry Pi 4 × 3	Physical IoT edge nodes (Phase 4)
IoT router	TP-Link + OpenWRT	Programmable gateway (Phase 4)
Language	Python 3.12.7	All components
OS	Windows 11	Development machine

SECTION 5 — CURRENT STATE (what is done right now)

COMPLETED — Phase 0

- Python 3.12.7 verified
- VS Code set up
- All packages installed: stable-baselines3, gymnasium, torch, numpy, matplotlib, requests
- CartPole RL test passed — RL stack confirmed working

COMPLETED — Phase 1

- ghostnet_env.py built — custom Gymnasium environment simulating hospital network
- 10-dimension state space defined (cloud IP, open ports, API, IoT IP, MQTT, CVE, Shodan, traffic, recon, time)
- 6-action mutation space defined
- TADR reward formula implemented
- PPO agent trained — 100,000 steps
- Best reward achieved: 159.8 out of 190 maximum = 84% optimal efficiency
- Best model saved: best_model/best_model.zip
- Training curve plotted and saved: ghostnet_results.png (Figure 1 for paper)
- Explained variance rose from 0.003 to 0.415 — agent learned environment dynamics

COMPLETED — Phase 2 (partial)

- threat_feeds.py built — connects to NIST NVD CVE API
- SSL issue on Windows Python 3.12 resolved using verify=False + urllib3.disable_warnings
- Live CVE test passed — pulled real CVEs from NIST this week
    - CVE-2026-35563 CVSS: 8.5
    - CVE-2026-27788 CVSS: 7.8
    - CVE-2026-32325 CVSS: 7.8
    - Normalized score: 0.85 — HIGH threat level
- ghostnet_env_v2.py built — upgraded environment with live CVE injection
- Shodan score: simulated (0.333-0.488) — real key optional, not blocking

PENDING — Phase 2 (one step remaining)

- step6_train_phase2.py NOT yet run
- Need to retrain agent with live CVE data using ghostnet_env_v2.py
- Command to run: python step6_train_phase2.py
- Expected output: ghostnet_v2.zip saved, reward similar to Phase 1

SECTION 6 — ALL FILES IN YOUR GHOSTNET FOLDER RIGHT NOW

```
C:\Users\SANGAVI\Documents\project\ghostnet\
|
├── ghostnet_env.py          Phase 1 core — hospital network simulator
├── ghostnet_env_v2.py       Phase 2 core — upgraded with live CVE
├── threat_feeds.py          Phase 2 core — NIST CVE API connector
|
├── step1_test_rl.py         Phase 0 — CartPole RL verification
├── step2_test_env.py        Phase 1 — environment test
├── step3_train_phase1.py    Phase 1 — PPO training (simulation)
├── step4_plot_results.py    Phase 1 — training curve graph
├── step5_test_threat_feed.py Phase 2 — CVE API test
├── step6_train_phase2.py    Phase 2 — PPO training (live CVE) ← NOT RUN YET
├── step7_final_demo.py      Phase 2 — live presentation demo
|
├── ghostnet_v1.zip          Trained agent Phase 1 (saved)
├── best_model/
|   └── best_model.zip       Best performing agent (use for demo)
├── logs/
|   ├── monitor.csv          Episode reward log
|   └── evaluations.npz      Eval checkpoints (used for graph)
|
├── ghostnet_results.png     Training curve — Figure 1 for paper
├── test_rl.py               Early CartPole test (ignore)
└── train.py                 Old train script (replaced by step3/step6)
```

SECTION 7 — NEXT STEPS IN ORDER

Immediate (today — 10 minutes)

```
1. Run: python step6_train_phase2.py
```

```
   Creates ghostnet_v2.zip — LTSA-trained agent
   Phase 2 fully complete after this

2. Run: python step7_final_demo.py
   Shows 20 live mutation decisions
   This is what you present in competitions
```

Phase 3 (needs AWS account — no credit card workaround)

```
Option A: Ask family member for card (no charges on free tier)
Option B: Google Cloud for Students — $300 free credit, student Gmail
Option C: Skip cloud for now, present simulation as proof of concept

What Phase 3 adds:
- cloud_mutator.py using boto3 to make real AWS Security Group changes
- Agent action 1 (close_port) triggers real AWS API call
- Makes project production-grade not just simulation
```

Phase 4 (hardware — ~₹18,000-20,000)

```
Buy only after Phase 3 working:
- 3x Raspberry Pi 4 (4GB) — ₹4,000 each
- 1x TP-Link Archer C7 / GL.iNet router (OpenWRT compatible) — ₹2,500
- 1x 8-port Ethernet switch — ₹900
- 3x 32GB MicroSD + accessories — ₹1,500
- 3x DHT22 temperature sensors — ₹600

Check router compatibility before buying:
https://openwrt.org/toh/start

What Phase 4 adds:
- Real IoT devices sending MQTT data
- GhostNet mutates physical router firewall rules
- DUAS fully implemented in hardware
```

Phase 5 (CALDERA red agent — free, no hardware)

```
Install CALDERA: git clone https://github.com/mitre/caldera.git
Run: docker run -p 8888:8888 caldera

What Phase 5 adds:
- Red agent launches real Nmap scans against testbed
```

```
- Blue agent (GhostNet) mutates in response
- Nash equilibrium convergence curve plotted
- This is ARAC implemented — journal-grade contribution
```

Phase 6 (paper write-up)

```
Write sections in this order (never intro first):
1. Experiments and Results — you have the data
2. System Design — you built it
3. Related Work — 5 papers identified
4. Methodology — MDP formulation
5. Abstract — summarise what you built
6. Introduction — very last

Target conference: IEEE INFOCOM 2027
Alternative: IEEE Access (faster, open access)
India option: COMSNETS, ANTS
```

SECTION 8 — COST SUMMARY

Item	Cost	Status
Python + VS Code	Free	Done
All Python packages	Free	Done
NIST CVE API	Free forever	Working
Shodan academic key	Free (student email)	Optional
AWS Free Tier	Free 12 months (needs card)	Pending
Google Cloud for Students	$300 free (student Gmail)	Alternative
MITRE CALDERA	Free open source	Phase 5
Raspberry Pi 4 × 3	₹12,000	Phase 4
OpenWRT router	₹2,500	Phase 4
Switch + accessories	₹2,000	Phase 4
Sensors + wires	₹600	Phase 4

Item	Cost	Status
Total hardware	₹17,100	Phase 4
Total software	₹0	All phases

SECTION 9 — LICENSES (no permission needed)

Tool	License	Safe to use?
stable-baselines3	MIT	Yes — commercial and academic
Gymnasium	MIT	Yes
MITRE CALDERA	Apache 2.0	Yes — only on your own testbed
MITRE ATT&CK	CC BY 4.0	Yes — cite MITRE in paper
NIST NVD CVE API	Public domain	Yes — free forever
Terraform CDK	MPL 2.0	Yes — free non-commercial
AWS Free Tier	AWS ToS	Yes — stay within free limits
OpenWRT	GPL v2	Yes — free to install
Mosquitto MQTT	EPL/EDL	Yes — free open source
PyTorch	BSD	Yes

LEGAL WARNING: Run CALDERA attacks ONLY on your own testbed. Never on any network you do not own. This is a legal requirement under IT Act India and CFAA internationally.

SECTION 10 — ATTACKS GHOSTNET PREVENTS (14 total)

Reconnaissance: Port scanning, API enumeration, MQTT topic hijacking, IoT fingerprinting, zero-day exploitation window
Lateral movement: IoT-to-cloud credential theft, cloud-to-IoT command injection, east-west subnet movement
Injection: Drug dose tampering via forged MQTT command, patient vitals spoofing

Ransomware / APT: Ransomware deployment (AIIMS pattern), APT long-term infiltration
DoS: Targeted DoS against hospital cloud APIs
Credential: API key and session token replay attacks

Does NOT prevent: Physical tampering, insider threats, phishing, sub-60-second automated attacks.

SECTION 11 — KEY NUMBERS TO REMEMBER

```
Best reward achieved      : 159.8
Maximum possible reward   : ~190
Defense efficiency        : 84.1%
Training steps            : 100,000
State dimensions          : 10
Mutation actions          : 6
Live CVE score today      : 0.85 (HIGH threat — real data)
Episodes per training     : ~500
Hardware cost (Phase 4)   : ₹17,100
Software cost (all)       : ₹0
```

SECTION 12 — COMPETITION OPENING (memorise this)

"In November 2022, AIIMS Delhi — India's most advanced hospital — was shut down for 15 days by ransomware. 40 million patient records were exposed. The attackers spent weeks silently mapping the network before they struck. Nothing stopped them during those weeks because the network never moved.

GhostNet solves exactly that. It is an AI agent that continuously changes the shape of the hospital network — rotating IPs, closing ports, changing API paths, shifting MQTT channels — across both the cloud and IoT layers simultaneously, guided by live threat intelligence, without ever disrupting patient care.

The University of Toronto published a paper in May 2025 calling RL-driven, context-aware MTD an open research problem. GhostNet is the answer."

SECTION 13 — VIVA DEFENCE ANSWERS

Q: Why do you need to secure both cloud and IoT?
A: They are not two systems. They are one pipeline. IoT collects, cloud processes, decisions

flow back. Every major hospital breach — AIIMS 2022, Target 2013, Oldsmar 2021 — crossed from one to the other. Securing only one is like locking the front door but leaving the back door open with a sign on it.

Q: What companies are already doing this?
A: Palo Alto does Zero Trust for medical IoT — reactive, not proactive. Claroty does device visibility — tells you when something is wrong, doesn't prevent the mapping. Armis does asset intelligence — passive only. None implement Moving Target Defense. None have a unified cloud+IoT agent.

Q: How is your reward function different?
A: Every existing RL-MTD paper uses a single objective — maximise attacker disruption. TADR uses two objectives simultaneously — maximise attacker disruption AND minimise legitimate traffic disruption. The second term is critical for hospitals because a mutation that disconnects an active infusion pump is worse than no mutation at all.

Q: Is this hackable?
A: Every system is hackable. GhostNet's vulnerabilities are: timing pattern prediction by a patient attacker, mutation registry compromise, and attacks that complete before the first mutation cycle. These are significantly harder to exploit than attacking a static network. The bar goes from any script kiddie to nation-state level capability. That difference saves lives.

Q: No existing system solves this — how can you be sure?
A: Precisely: no existing system simultaneously addresses reconnaissance-phase attacks across both hospital IoT and cloud infrastructure under a single unified intelligent agent with traffic-aware mutation timing. Morphisec does cloud MTD. Armis does IoT security. Neither does both under one RL policy. The VIT Chennai paper uses rule-based MTD on AWS. The U of Toronto paper names our exact contribution as an open problem in May 2025.

SECTION 14 — COMPANIES TO APPROACH

Apollo Hospitals, Narayana Health, Manipal Hospitals — large IoT-connected hospital chains in India
Philips Healthcare India, GE Healthcare India, Siemens Healthineers — medical device makers
CERT-In — Government of India cybersecurity agency
MeitY — Ministry of Electronics and IT (grant funding available)
NIC — National Informatics Centre (runs AIIMS digital infrastructure)
Tata 1mg, Practo — health-tech startups with cloud+IoT infrastructure

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END OF HANDOFF REPORT